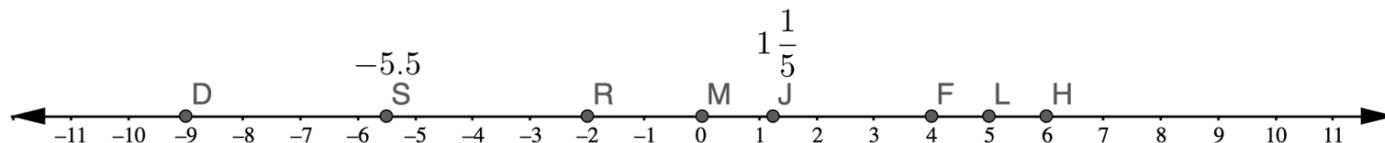


Grade 6 Accelerated
Unit 3
Lesson 1 Homework

Name Answer Key
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A number line with points D , S , R , M , J , F , L , and H is shown.



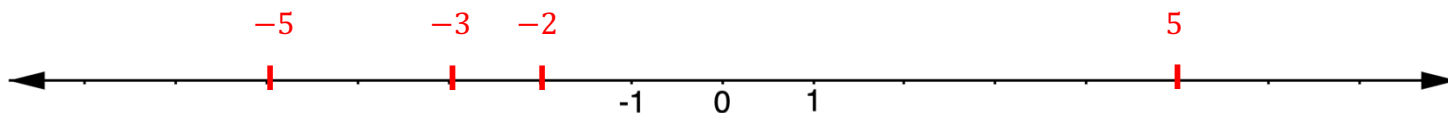
1. Complete the table by identifying which points belong in each category. Some points may be used more than once and some may not be used at all.

Natural Numbers	Whole Numbers	Integers	Rational Numbers
$F L H$	$M F L H$	$D R M$ $F L H$	$D S R$ $M J F$ $L H$

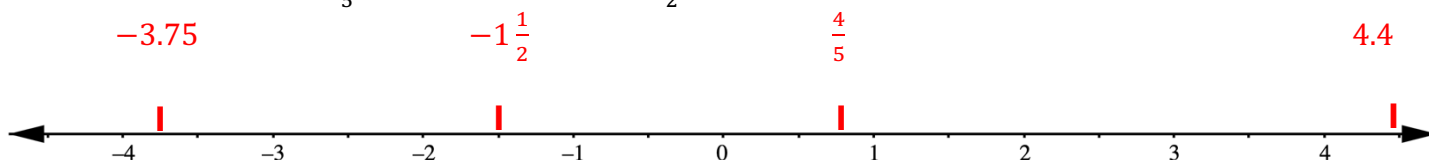
2. Complete the table by identifying which points belong in each category. Some points may be used more than once and some may not be used at all.

Positive Numbers	Negative Numbers
$J F L H$	$R S D$

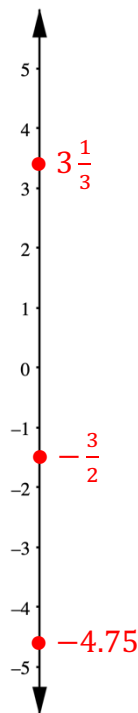
3. Plot the integers -2 , -3 , 5 , and -5 on the number line.



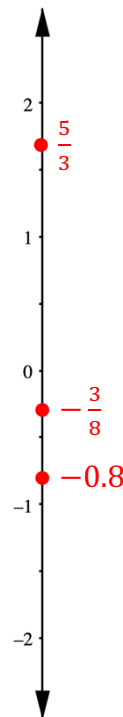
4. Plot the numbers $\frac{4}{5}$, -3.75 , 4.4 , and $-1\frac{1}{2}$ on the number line.



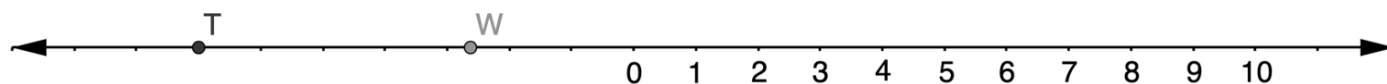
5. Plot the numbers $-\frac{3}{2}$, $3\frac{1}{3}$, and -4.75 .



6. Plot the numbers $\frac{5}{3}$, $-\frac{3}{8}$, and -0.8 .



A number line with points T and W is shown.



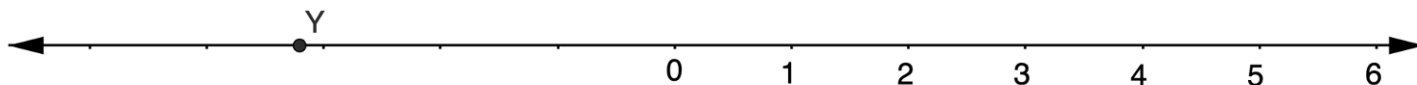
7. Explain how to estimate the value of T .

First count by integers to find which value it is closest to and T appears to be at -7 .

8. Explain how to estimate the value of W .

First count by integers to find which values it appears to be between. W is between -2 and -3 , it is past the halfway point so a good estimate would be -2.75 .

Mariah and Jackson both estimated the value of Y on the number line shown.



9. Mariah says the value of Y is about -4.8 . Explain whether you agree with Mariah.

I disagree it is just past -3 not just before -5 .

10. Jackson says the value of Y is about $-3\frac{1}{2}$. Explain whether you agree with Jackson.

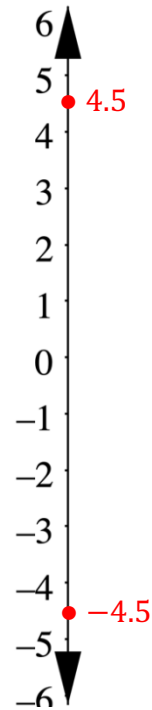
I disagree, Jackson is closer than Mariah but the value Y is closer to -3 than the halfway point, as Jackson suggests.

Use the vertical number line to answer the following problems.

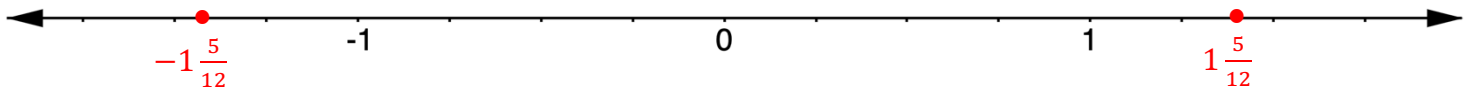
1. Plot -4.5 and its opposite.

2. Which is larger, -4.5 or its opposite?

Its opposite of 4.5 is larger.



Use the horizontal number line to answer the following problems.

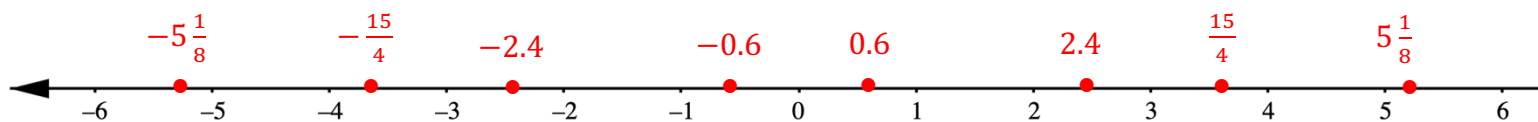


3. Plot $1\frac{5}{12}$ and its opposite.

4. Which is larger, $1\frac{5}{12}$ or its opposite?

$1\frac{5}{12}$ is larger.

5. Plot and label each value and its opposite on the number line: -2.4 , $5\frac{1}{8}$, 0.6 , $-\frac{15}{4}$



Complete each comparison statement for the pairs of numbers you plotted in question 5. Use $<$ or $>$ in the first blank and the given number's opposite in the second blank.

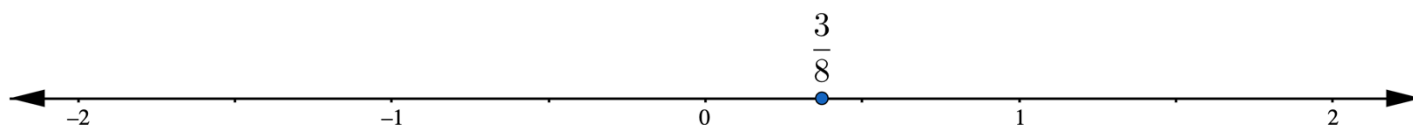
6. $-2.4 < 2.4$

7. $5\frac{1}{8} > -5\frac{1}{8}$

8. $0.6 > -0.6$

9. $-\frac{15}{4} < \frac{15}{4}$

10. A number line with the value $\frac{3}{8}$ plotted is shown.



Kyree said that the opposite of $\frac{3}{8}$ is $\frac{8}{3}$ and so $\frac{3}{8}$ is less than its opposite. Explain whether you agree with Kyree.

I disagree. $\frac{8}{3}$ is the reciprocal of $\frac{3}{8}$ not the opposite. The opposite is $-\frac{3}{8}$ of $\frac{3}{8}$.

Grade 6 Accelerated
Unit 3
Lesson 3 Homework

Name Answer Key
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Explain why each inequality statement is true or false.

1. $-\frac{1}{3} > -\frac{3}{2}$

True, $-\frac{1}{3}$ is closer to zero and farther right on a number line than $-\frac{3}{2}$ is.

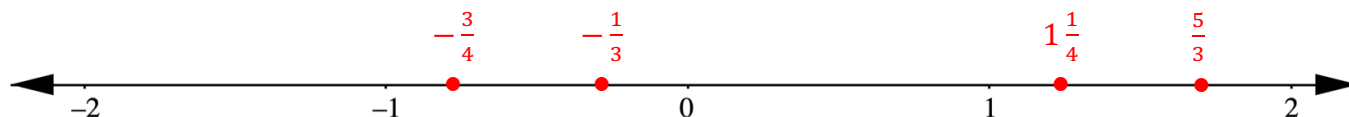
2. $-2\frac{2}{5} > -\frac{9}{4}$

False, $-2\frac{2}{5}$ is less than $-\frac{9}{4}$ as it is farther left on the number line than $-\frac{9}{4}$ is.

A set of rational numbers is shown.

$$-\frac{3}{4}, \quad 1\frac{1}{4}, \quad \frac{5}{3}, \quad -\frac{1}{3}$$

3. Plot each of the values on the number line. Label each point with its numeric value.



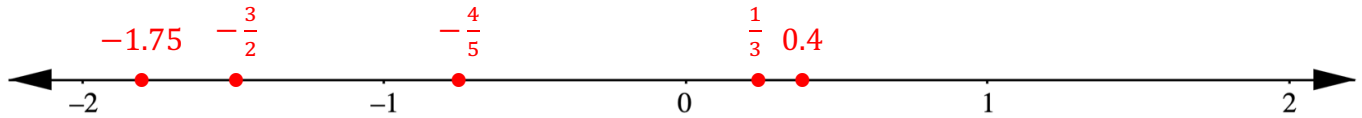
4. Complete the inequality statement to show the relationship between the values.

$$-\frac{3}{4} < -\frac{1}{3} < 1\frac{1}{4} < \frac{5}{3}$$

A set of rational numbers is shown.

$$-\frac{3}{2}, \quad -1.75, \quad \frac{1}{3}, \quad -\frac{4}{5}, \quad 0.4$$

5. Plot each of the values on the number line. Label each point with its numeric value.



6. Complete the inequality statement to show the relationship between the values.

$$-1.75 < -\frac{3}{2} < -\frac{4}{5} < \frac{1}{3} < 0.4$$

Complete each statement using $<$, $>$, or $=$.

7. $-\frac{5}{2} < -2\frac{1}{3}$

8. $-0.6 < \frac{3}{5}$

9. Order the values from least to greatest.

$$-3\frac{2}{3}, \quad -\frac{7}{2}, \quad \frac{9}{10}, \quad -\frac{9}{10}, \quad 0, \quad 1\frac{1}{4}$$

$$-3\frac{2}{3}, -\frac{7}{2}, -\frac{9}{10}, 0, \frac{9}{10}, 1\frac{1}{4}$$

10. Order the values from least to greatest.

$$-2.8, \quad -\frac{8}{3}, \quad \frac{11}{4}, \quad -2.25, \quad 2\frac{1}{5}$$

$$-2.8, -\frac{8}{3}, -2.25, 2\frac{1}{5}, \frac{11}{4}$$

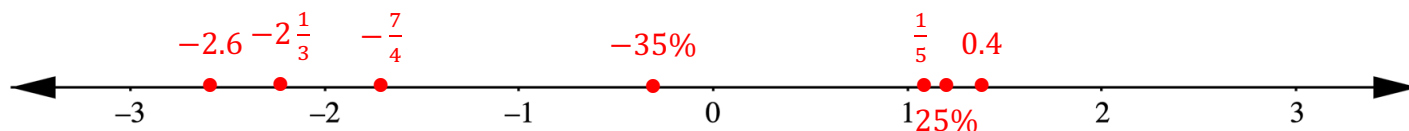
Grade 6 Accelerated
Unit 3
Lesson 4 Homework

Name Answer Key
Date _____
Period _____

A set of rational numbers is shown.

$$\begin{array}{ccccccc} -\frac{7}{4}, & -2.6, & 25\%, & -2\frac{1}{3}, & \frac{1}{5}, & -35\%, & 0.4 \\ = -1.75 & & = 0.25 & = -2.\bar{3} & = 0.2 & = -0.35 & \end{array}$$

1. Plot each of the values on the number line. Label each point with its numeric value.



2. Order the values from least to greatest.

$$-2.6, -2\frac{1}{3}, -\frac{7}{4}, -35\%, \frac{1}{5}, 25\%, 0.4$$

Complete each statement using $<$, $>$, or $=$.

3. $-4.825 < -4\frac{4}{5}$

4. $63\% < \frac{2}{3}$

5. $-28\% < 0.28$

6. $-175\% = -\frac{7}{4}$

7. $-12.412 < -12\frac{3}{8}$

8. $3.5\% < \frac{7}{2}$

9. Order the values from least to greatest.

$$-1\frac{7}{8}, \quad 78\%, \quad -\frac{7}{4}, \quad -1.586, \quad 0.77, \quad \frac{7}{10}$$

$$-1\frac{7}{8}, -\frac{7}{4}, -1.586, \frac{7}{10}, 0.77, 78\%$$

10. Dion wrote the inequality statement:

$$-3.125 < -3\% < -\frac{13}{5}$$

Explain whether or not you agree with Dion.

$$\begin{array}{l} -3.125 \\ -.03 \\ -2.6 \end{array}$$

I disagree although -3.125 is the smallest number $-\frac{13}{5}$ is smaller than -3% .

Grade 6 Accelerated
Unit 3
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

Write a value that represents each description.

1. A temperature drop of 14°
 -14°
2. A withdrawal of \$29.97 from a checking account
 -29.97
3. A gain of 4 yards in a football game
 $+4$ or 4
4. An elevation of 5,836 feet above sea level
 $+5,836$ or $5,836$

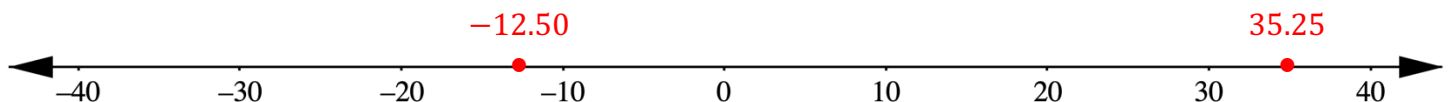
Marcy and Anya are comparing the money in their bank accounts. Marcy just opened her account with a deposit of \$35.25. Anya owes the bank \$12.50.

5. Write a rational number that represents each person's account balance.

Marcy: 35.25

Anya: -12.50

6. Plot both rational numbers on the number line.



7. What does 0 represent in this context?

0 represents no money in the bank account.

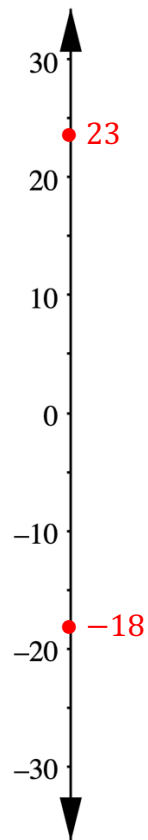
A scientist is tracking animals in Blue Spring State Park. She notices that there is a pelican flying 23 feet above sea level and a manatee swimming 18 feet below sea level.

8. Write a rational number that represents each animal's elevation.

Pelican: 23

Manatee: -18

9. Plot both rational numbers on the number line.



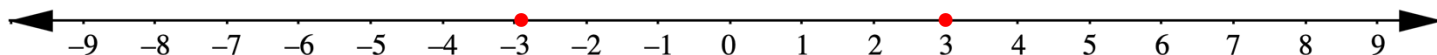
10. What does 0 represent in this context?

In this context 0 represents at sea level.

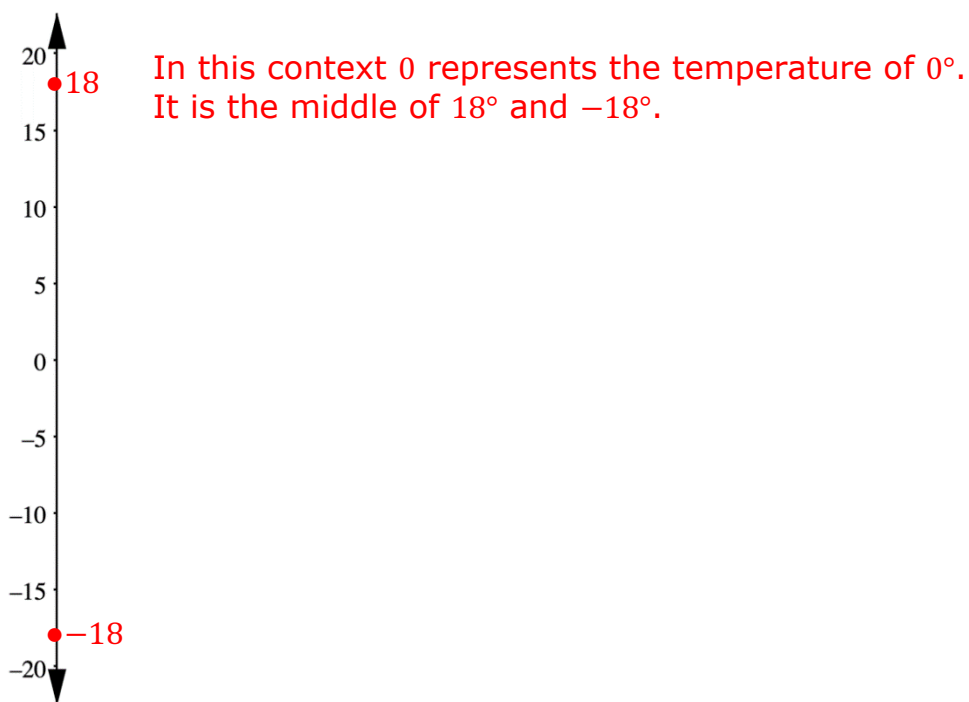
Grade 6 Accelerated
Unit 3
Lesson 6 Homework

Name Answer Key
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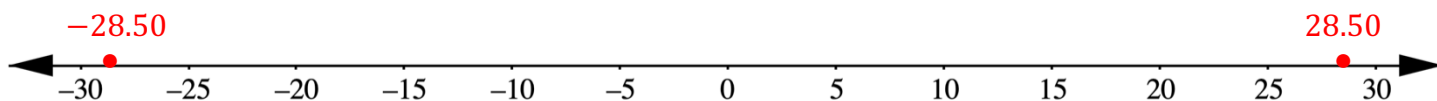
1. Plot -3 and its opposite on the number line.



2. On the number line, plot a temperature of 18° below zero and its opposite. What is the meaning of 0 in this context?



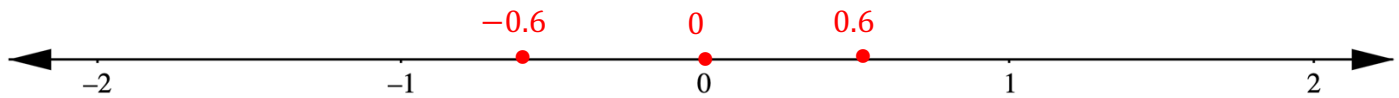
3. Plot a withdrawal of \$28.50 and a deposit of the same amount on the number line. What is the meaning of 0 in this context?



In this context 0 means no deposits or withdrawals on the account.

From Isaac's house, if he walks 0.6 miles West, he'll reach the library, and if he walks 0.6 miles East, he'll reach the park.

4. Model all three locations on the number line.



5. What does 0 represent in this situation?

0 represents Isaac's house.

6. From his house, is it farther for Isaac to go to the library or to the park?

It is equal distance, just opposite directions.

Complete the table for each description.

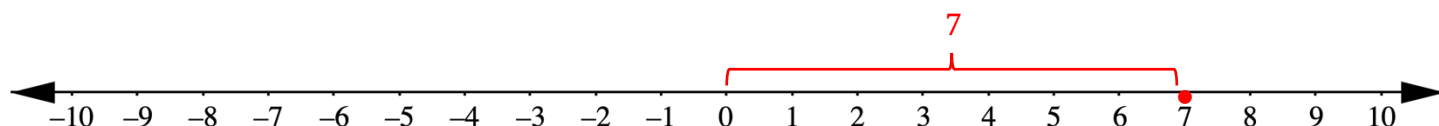
	Description	Rational Number	Opposite	Meaning of Opposite	Meaning of 0
7.	A deposit of \$182.37	182.37	-182.37	Withdrawal of 182.37.	No withdrawal or deposit in the account.
8.	A gain of 5 yards	5	-5	Loss of 5 yards.	No loss or gain of yards.
9.	11,729 feet below sea level	-11,729	11,729	11,729 ft above sea level.	At sea level.
10.	A temperature increase of 2.8°	2.8	-2.8	A temperature drop of 2.8°.	0 degrees, no loss or gains in temperature.

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Unit 3
Lesson 7 Homework

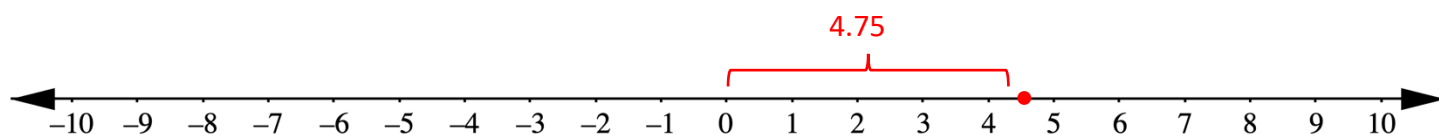
Name Answer Key
Date _____
Period _____

Complete each statement and model it on the number line.

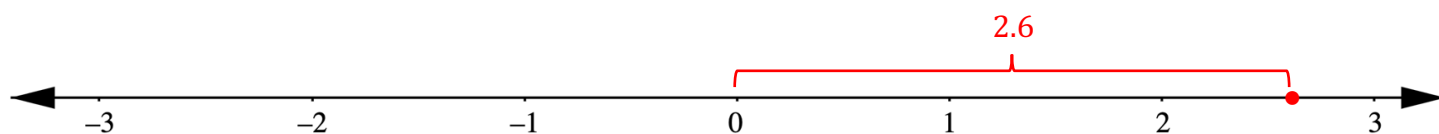
1. $|-7| = \underline{7}$



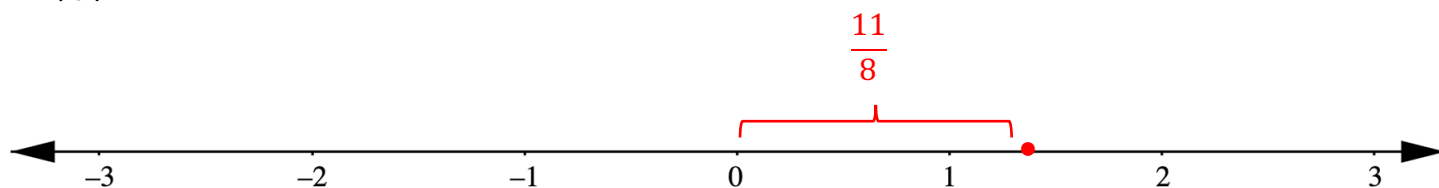
2. $|4.75| = \underline{4.75}$



3. $|-2.6| = \underline{2.6}$



4. $\left|\frac{11}{8}\right| = \underline{\frac{11}{8}}$



5. Sarita claims that $\left|5\frac{1}{3}\right| = -5\frac{1}{3}$. Explain whether you agree with Sarita.

I disagree with Sarita. The absolute value is how far away (distance) a number is from zero so $\left|5\frac{1}{3}\right|$ is $5\frac{1}{3}$ units away from zero.

Gerald received a bank statement that said his balance was $-\$72.85$.

6. What does it mean for Gerald's account balance to be negative?

It means he has spent more money than he has in his account.

7. What does the absolute value of Gerald's account balance mean in this context?

The absolute value of this balance would mean he has a balance of $\$72.85$ in his account.

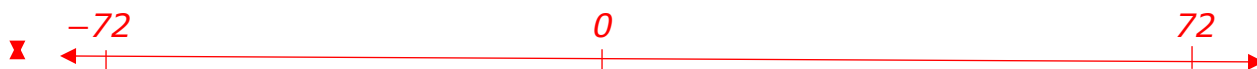
8. The average elevation of the Florida Keys is 3.2 feet. What does the absolute value of the elevation mean in this context?

It means the elevation on average is 3.2 feet above sea level.

9. A bank account has a balance of $\$220$. Describe the meaning of each value in the context of transactions on the bank account.

Transaction	Interpretation
$\$53.45$	Gain of 53.45, he now has 273.45 in his account. (deposit)
$-\$18.99$	Loss of 18.99 (withdrawal), he now has 254.46 in his account.
$\$ 150 $	Gain (deposit) of 150, he now has 404.46 in his account.
$\$ 72.95 $	Gain (deposit) of 72.95, he now has 477.41 in his account.

10. The average high temperature in Anchorage, Alaska in February is 72° from 0°C . What could be the average high temperature in Anchorage, Alaska in February?



If it can be from 0 to 72 that means it could be either direction so the average can be -72° or 72° .

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Unit 3
Lesson 8 Homework

Name Answer Key
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Use the symbols $<$, $>$, and $=$ to compare each pair of values.

1. $|-8|$ $>$ $|6|$
 8 6

2. 2 $<$ $\left|-\frac{7}{3}\right|$
 2 $\frac{7}{3}$
 2 $2\frac{1}{3}$

3. $|-9.4|$ $=$ $|9.4|$
 9.4 9.4

4. $|-64|$ $>$ -13
 64 -13

5. $\left|\frac{5}{4}\right|$ $=$ 1.25
 1.25 1.25

6. $|18.3|$ $>$ -18.3
 18.3 -18.3

7. Angelina started a company making and selling bracelets. Her expenses were \$142. Her revenue was \$118. Write a value that could be used to describe each part of Angelina's finances.

Angelina's Finances	Value
Weekly Expenses	-142
Weekly Revenue	118

Ravi and Celine are scuba diving off the Florida Keys. Ravi's elevation is currently -27 feet, and Celine's elevation is -32 feet.

8. Complete each statement.

Ravi is currently 27 feet ☒ below sea level.

Celine is currently 32 feet ☒ below sea level.

9. Use an inequality symbol, $>$ or $<$, to compare the divers' elevations.

$$-27 \text{ } > \text{ } -32$$

10. Use an inequality symbol, $>$ or $<$, to compare the divers' distances from zero.

$$|-27| \text{ } < \text{ } |-32|$$

Grade 6 Accelerated
Unit 3
Lesson 9 Homework

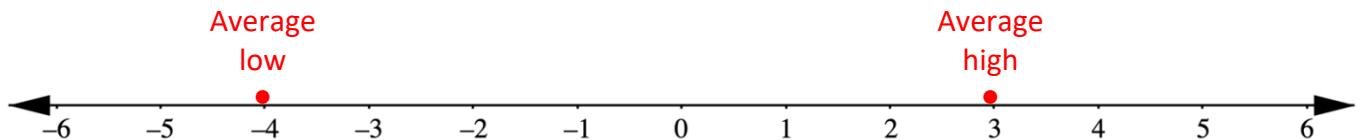
Name Answer Key
Date _____
Period _____

The freezing point is 0°C . In December in Chicago, Illinois, the average high temperature is 3°C above freezing and the average low temperature is 4°C below freezing.

1. Write the values that represent the average high temperature and the average low temperature in Chicago in December.

Description	Value
Average high temperature	3
Average low temperature	-4

2. Model the two values on the number line.



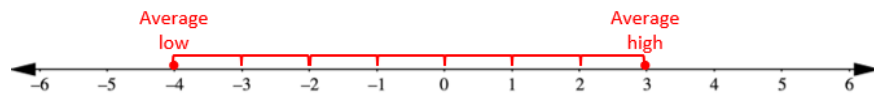
3. Write a mathematical statement using absolute value that could be used to find the difference between the average high and average low temperatures.

$$|-4| + |3|$$

4. What is the difference between the average high temperature and the average low temperature?

$$|-4| + |3|$$

$$4 + 3 = 7$$



5. Write a mathematical statement using absolute value that could be used to determine how much farther from freezing the average low temperature is than the average high temperature.

$$|-4| - |3|$$

6. How much farther from freezing is the low temperature than the high temperature?

$$|-4| - |3|$$

$$4 - 3 = 1$$

Allan withdrew money from his bank account three times last week. Each time, the transaction on his bank statement said $-\$40$.

7. Write a mathematical statement using absolute value that could be used to determine the total amount of money Allan withdrew last week.

$$3 \times |-40|$$

8. What is the total amount of money Allan withdrew last week?

$$\begin{aligned} 3 \times |-40| & \quad \text{Allan withdrew \$120 last week.} \\ 3 \times 40 & \\ = 120 & \end{aligned}$$

A dolphin and a turtle are both swimming. The dolphin is 6 feet below sea level, and the turtle is 31 feet below sea level.

9. Write a mathematical statement using absolute value that could be used to determine how much farther from sea level the turtle is than the dolphin.

$$|-31| - |-6|$$

10. How much farther from sea level is the turtle than the dolphin?

$$\begin{aligned} & |-31| - |-6| \\ & 31 - 6 \\ & = 25 \end{aligned}$$

The turtle is 25 ft further below sea level than the dolphin is.